

EE 210L: Digital Circuits and Logic Design Lab

Lab 2: Boolean Simplification and K-Maps

Name: _____

Date: _____

Partner(s): _____

Section: _____

Show every step of your minimization, not just the final answer. Draw your groupings directly on the K-maps.

Part 4.1: Majority Voter

Truth Table

A	B	C	M	Minterm (if M = 1)
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

Unminimized SOP Expression

M = _____

K-Map

Fill in each cell and circle your groups.

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0				
A = 1				

Minimized Expression

M = _____

Part 4.2: Four-Variable Function

$F(A,B,C,D) = \sum m(0, 1, 2, 3, 5, 7, 8, 9, 10, 11)$

K-Map

Fill in each cell and circle your groups. One group covers eight cells.

	CD = 00	CD = 01	CD = 11	CD = 10
AB = 00				
AB = 01				
AB = 11				
AB = 10				

Minimized Expression

F = _____

Part 5.1: Simulation

Paste a screenshot of each simulated schematic in the boxes below.

Figure 1: Simulated majority voter (minimized).

Figure 2: Simulated four-variable function (minimized).

Did both simulations match your pre-lab truth tables on the first try? If not, what was wrong?

Part 5.2: Measured Majority Voter

A	B	C	Measured V_{out} (V)	Logic Output	Expected M
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

Part 5.3: Measured Four-Variable Function

A	B	C	D	Measured V_{out} (V)	Logic Output	Expected F
0	0	0	0			
0	0	0	1			
0	0	1	0			
0	0	1	1			
0	1	0	0			
0	1	0	1			
0	1	1	0			
0	1	1	1			
1	0	0	0			
1	0	0	1			
1	0	1	0			
1	0	1	1			
1	1	0	0			
1	1	0	1			
1	1	1	0			
1	1	1	1			

Part 5.4: Cost Comparison

Count 2-input gates. A 3-input AND built from 2-input gates counts as two gates. Count inverters separately.

Function	Version	Gates	Inverters	IC packages
Majority voter	Unminimized SOP			
Majority voter	Minimized			
F(A,B,C,D)	Unminimized SOP			
F(A,B,C,D)	Minimized			

Discussion and Conclusions

Address at least: how much hardware minimization saved you, which K-map grouping was hardest to spot and why, and whether simulating first saved you time at the bench.
